

SAMUDRA

SAR Attribution of **M**arine **U**nauthorised **D**ischarge, **R**anking and **A**ssessment

Oil-Spill Detection, Drift Reconstruction and Vessel Attribution Engine
Smart India Hackathon — Problem Statement SIH26143 (NTRO)

Complete project report: aim, architecture, algorithms, data, implementation, user interface, verification results, limitations and how to run it.

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1. Executive summary

Detecting an oil slick from satellite radar is a solved problem — services such as EMSA CleanSeaNet and SkyTruth Cerulean already do it. **Naming the ship that most likely released it, with a defensible number attached to that claim, is not.** SAMUDRA addresses that second problem.

Given a SAR (synthetic-aperture radar) scene showing a slick, the surrounding ship traffic (AIS) and a wind and ocean-current field, SAMUDRA:

- drifts the observed slick **backwards in time** to find where the oil could have come from (the origin envelope);
- **filters** the vessel traffic down to only the ships that were inside that envelope at the right time;
- for every surviving ship, **simulates a discharge** along its own track and drifts it forward to the moment the satellite looked, then compares the simulated slick with the real one;
- **ranks** the candidates with an attribution score that combines that geometric match with contextual factors (vessel type, AIS integrity, behaviour, AIS gaps);
- **forecasts** where the slick will go over the next 24, 48 and 72 hours and whether it reaches the coast;
- writes a **prosecution-oriented evidence dossier** (PDF) protected by a SHA-256 hash chain, and composes a tiered alert for the Indian Coast Guard.

On synthetic scenarios with a planted culprit, the engine ranks the correct vessel first on scenarios it was never tuned against, with release-time errors of minutes and release-position errors of 1-2 km. Real ISRO EOS-04 SAR imagery is ingested end to end, and real NOAA AIS has been used to test the engine against genuine traffic.

The output is always a **ranked list of candidate vessels**, never a single accusation: the system is decision-support evidence and does not establish legal responsibility.

2. The problem and the aim

2.1 The problem

Illegal operational discharges of oily waste from ships (bilge water, tank washings) are a chronic source of marine pollution. They are usually small, deliberate and done at night or far offshore, where no one is watching. A satellite may later reveal the slick, but by then the vessel has sailed on. Existing systems stop at detection or, at best, point at the nearest ship track — which is often wrong, because oil drifts with wind and current for hours before it is seen, and because AIS (the ship self-reporting system) can be switched off or spoofed.

2.2 What the problem statement asks for

Requirement in the brief	Where SAMUDRA delivers it
(a) Detect the spill and characterise its geometry and age	Detection (DeepLabv3+), polygonisation, geometric features, slick-age estimate
(b) Hindcast to the origin and forecast future spread	Reverse particle drift -> origin envelope; forward drift at +24/48/72 h with coastline impact
(c) Attribute using proximity, trajectory and behavioural anomalies; filter irrelevant traffic	Candidate pruning funnel, discharge-hypothesis simulation, trust and behaviour scoring, ranked posteriors
Suitable visual interface	Four web dashboards including the new investigator view
Disaster-management linkage	NOS-DCP-tiered Coast Guard alert with distribution list and GeoJSON export

2.3 The aim

To build an **attribution engine** that answers *which vessel*, with a defensible likelihood, against an adversary who may be actively evading detection — and whose output is an evidence package suitable for the Indian Coast Guard under the National Oil Spill Disaster Contingency Plan (NOS-DCP) and MARPOL Annex I. Because satellite revisit makes real-time interception impossible, the achievable goal is **deterrence through reliable attribution**, and the project states that openly.

3. What SAMUDRA is — and is not

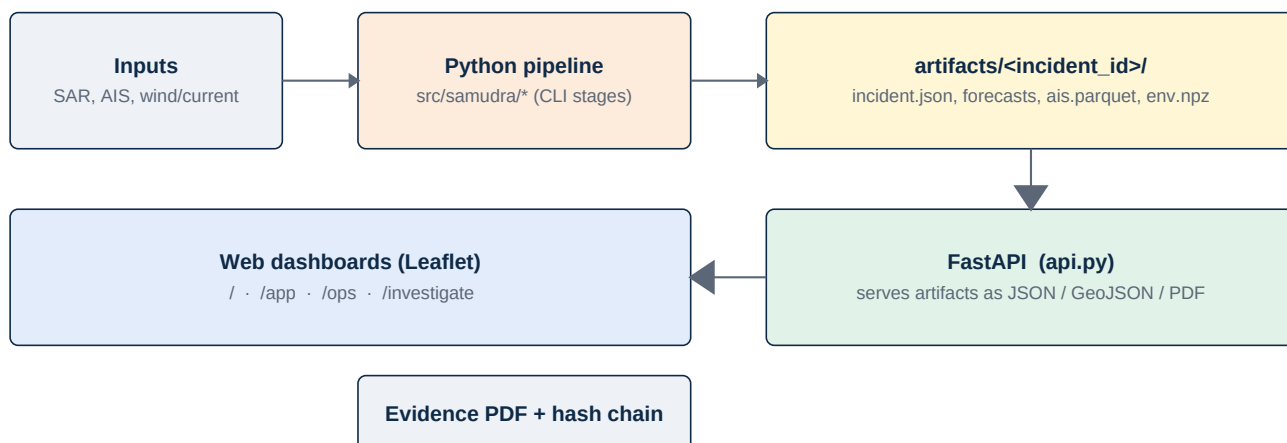
SAMUDRA is not "another oil-spill detector". Its five architectural commitments each close a gap in existing systems:

#	Gap in existing systems	SAMUDRA component
1	Detection is stateless — every scene judged in isolation	baseline/ — persistent per-cell backscatter memory, conditioned on wind
2	Attribution is nearest-track proximity	attribution/ — forward-simulation hypothesis testing
3	AIS is trusted or ignored, never scored	trust/ — adversarial credibility and behaviour scoring
4	Revisit gaps are dead time	ingest/ + satellite/ — opportunistic multi-sensor context (EOS-04, INSAT-3DS, EOS-06)
5	Irrelevant traffic is filtered silently	attribution/prune.py — named pruning stage with a reported funnel

What it deliberately does **not** claim:

- It does not confirm oil. SAR measures surface roughness; oil, biogenic films, rain cells and calm water all look dark. Every detection is reported as an *oil-like surface anomaly*.
- It does not accuse. Candidates are ranked by correlation; posteriors are relative scores, never a verdict.
- It does not intercept. Sentinel-1 / EOS-04 revisit over a given point is roughly 5-12 days, so the product is forensic attribution, not a live alarm.

4. System architecture (11 layers)



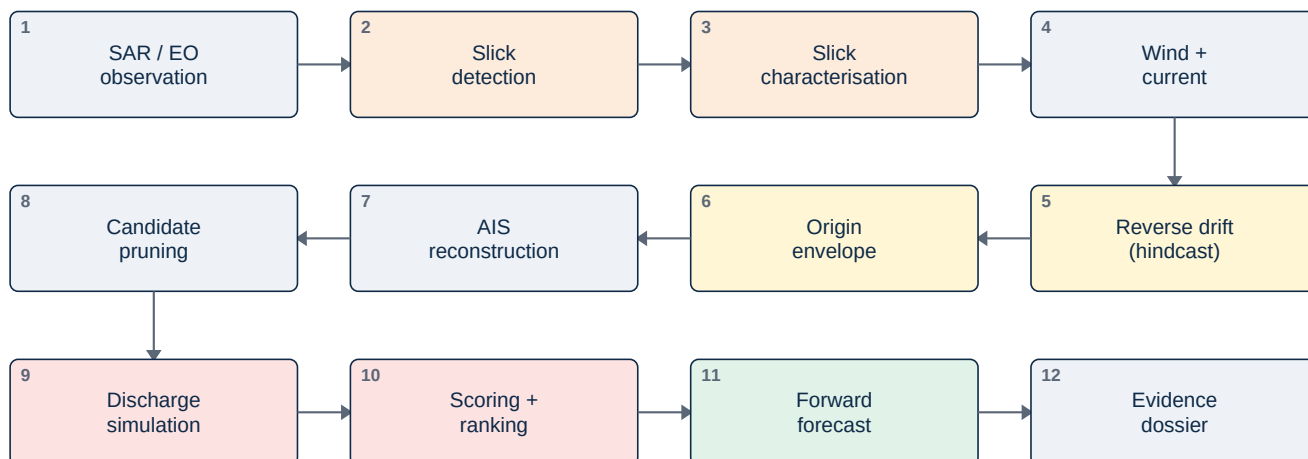
Every pipeline stage is a command-line entry point that reads from and writes to `artifacts/<incident_id>/`. The dashboards only read those files through the API, so what the screen shows and what the dossier records cannot diverge, and a demo replays deterministically from disk.

Layer	Name	Module(s)	Status
1	Data ingestion	ingest/ais.py, ingest/eos04.py, satellite/, monitor/	Built (real AIS, real EOS-04, MOSDAC search)
2	Preprocessing & baseline	detection/preprocess.py, baseline/anomaly.py	Built (baseline in fallback: 1 scene)
3	AI oil-slick detection	detection/segmenter.py, polygonize.py, fuse.py	Built, trained checkpoint
4	Vessel analysis (CFAR)	vessels/cfar.py	Built; SAR-AIS matching pending
5	AIS trust & behaviour	trust/score.py, trust/behaviour.py	Built
6	Environment & drift model	attribution/drift.py	Built
6b	Candidate pruning	attribution/prune.py	Built
7	Vessel attribution engine	attribution/hypothesis.py, rank.py	Built and verified

Layer	Name	Module(s)	Status
8	Impact & forecast	impact/forecast.py, coastline.py, landmask.py	Built
9	Investigator dashboard	api.py, web/	Built (4 views)
10	Evidence & integrity	evidence/report.py, integrity.py	Built
11	Dissemination & alerting	dissemination/alert.py	Built (composes, never transmits)

A feedback loop (connector A) is designed to carry confirmed incidents back into a vessel dossier database so that repeat offenders are weighted on future incidents. This is what would make the system improve over time; it is architected but not yet built (see section 16).

5. How it works, step by step



Step 1 — Observation (layer 1)

A SAR scene arrives: either a real ISRO EOS-04 product dropped into data/satellite/incoming/, a research archive chip, or a synthetic scene generated with known ground truth. For EOS-04, the ingest module verifies the product against itself: the published sigma0, gamma0 and beta0 calibration constants must satisfy their trigonometric identities (they hold to 0.0018 dB and 0.0000 dB), and the timezone of the scene-centre time is established by comparing modelled and published sun elevation (+7.50° as UTC vs the published +6.64°; -44° as IST). A wrong timezone would shift every back-projected release point by tens of kilometres.

Step 2 — Preprocessing and baseline (layer 2)

Land is masked (500 m coastal buffer), pixels outside the 3-10 m/s wind window are gated out (below 3 m/s the sea is glassy and everything looks like oil; above 10 m/s waves scrub the slick away), and the scene is tiled 512×512 with 64 px overlap. The baseline module keeps running statistics of backscatter per H3 cell and per wind bin, so a slick is judged against what that patch of sea normally looks like under those winds. It needs at least 15 scenes per cell; with fewer it falls back to a global estimate and says so.

Step 3 — Detection and characterisation (layer 3)

A DeepLabv3+ network with a ResNet-50 backbone segments oil-like pixels. The mask is turned into polygons and each polygon is characterised: area, perimeter, shape complexity (1.0 = circle), eccentricity, long-axis orientation, contrast and edge gradient. Confidence is fused from three signals:

```
confidence = cnn_oil_prob × sigmoid(baseline_anomaly_z - 1.5) × wind_gate_pass
```

Step 4 — Wind and current (layer 6)

A gridded wind and surface-current field over time (env . npz) drives all drift. In production this comes from EOS-06 scatterometer wind and ocean models; the demo AOIs use synthetic fields with realistic statistics.

Step 5 — Reverse drift / hindcast (layer 6)

Particles seeded in the observed slick are advected backwards in 5-minute steps:

```
v_particle = v_current + 0.03 × v_wind (rotated 15° clockwise, Coriolis)
spreading = random walk, K = 5 m²/s per particle per step
```

The result is not a point but a **space-time origin envelope**: one polygon per hour back (16 hours in the demos), each saying where the oil probably was at that time.

Step 6 — Candidate pruning (layer 6b)

Every AIS track is tested against every hourly envelope polygon, at that polygon's time and 30 minutes either side. A vessel survives only if it was inside the envelope *at the matching time* — passing the same water at a

different hour does not count. Survivors are ordered by how often they were inside while under way (a moored ship sits inside the envelope at every step and would otherwise dominate). The narrowing is reported as a funnel:

Stage	demo-001	kutch-exercise-001
Vessels reporting in scene	24	164
Inside origin envelope	3	3
Scored (discharge simulated)	3	3
Ranked	3	3

Step 7 — Discharge simulation (layer 7)

For each surviving vessel, release times are sampled every 30 minutes along its own track. A moving ship is a *line* source, so particles are released along its track over ~45 minutes, then drifted forward to acquisition time. The simulated slick is compared with the observed one on four measures: overlap (IoU), distance between centres, difference in long-axis orientation, and area ratio. The best hypothesis per vessel is kept.

Step 8 — Ranking (layer 7)

Geometry is turned into a relative likelihood across candidates and multiplied by contextual priors. See section 6.

Step 9 — Slick age

Age = acquisition time minus the winning release time, reported as a range across the top hypotheses. It is cross-checked against a Fay spreading law; that check depends on an assumed discharge volume, so it is a consistency check, not independent confirmation.

Step 10 — Forecast and coastline impact (layer 8)

The observed slick is drifted forward to +24, +48 and +72 hours with an uncertainty cone from particle spread. Each forecast polygon is intersected with the Natural Earth 1:10 m coastline to report affected shoreline length and an impact ETA; particles beach rather than advecting inland.

Step 11 — Evidence and integrity (layer 10)

A 12-section PDF dossier is generated: finding, detected slick, environment, funnel, slick age, ranked candidates with rationale, winning hypothesis map, trust and behaviour flags, forecast and impact, Coast Guard alert, MARPOL Annex I reference and chain of custody. SHA-256 hashes of all inputs and outputs are appended to `artifacts/audit.log`, each entry chained to the previous one — tampering with any file or the log is detected.

Step 12 — Dissemination (layer 11)

An alert is composed with NOS-DCP tiers (I local / II regional / III national) chosen from slick area and forecast shore-contact time, addressed to the Indian Coast Guard units that would respond (nearest station plus regional and, for Tier III, national HQ), with range and bearing. It is never transmitted — that is a deployment decision.

6. The scoring model explained

6.1 Geometric match score

```
match = 0.40 × IoU
        + 0.25 × exp(-centre_offset_km / 10)
        + 0.25 × cos(orientation_delta)
        + 0.10 × area_ratio           (min/max of the two areas)
```

Term	Plain meaning	Weight
Slick overlap (IoU)	How much of the observed slick the simulated discharge reproduces	40%
Centre proximity	How close the simulated slick's centre lands to the observed centre	25%
Course alignment	Whether the slick's long axis lies along the vessel's course — a moving ship lays an elongated slick	25%
Area similarity	How close the simulated area is to the observed area	10%

6.2 From geometry to attribution score

```
likelihood = softmax(match / T)   across candidates, T = 0.25
posterior  = likelihood × trust × behaviour × proximity × vessel_type
            × (1.8 if AIS gap at release) × (1 + 0.4 × prior_offences)
            -> renormalised so the candidates sum to 100%
```

Contextual factor	Values (config/weights.yaml)
Vessel type	tanker 1.40, bulk 1.25, cargo 1.10, container 1.00, tug 0.85, fishing 0.70
AIS trust	clean 1.0, minor flags 0.9, major flags 0.6, identity mismatch 1.3 (spoofing suggests intent)
Behaviour	loitering 1.25, slow-down 1.20, course deviation 1.15, night manoeuvring 1.10, AIS gap in AOI 1.35
Proximity	inside envelope 1.0, envelope edge 0.8
AIS gap at release	× 1.8
Prior confirmed offences	× (1 + 0.4 n) — always 0 until connector A is built

Why temperature 0.25? The match score lives in [0,1], so a plain softmax barely separates candidates ($\exp(0.90)/\exp(0.29) = 1.8\times$), and priors could overturn geometry: at $T = 1.0$ a vessel whose simulated slick was 86° misaligned ranked second on priors alone. $T = 0.25$ restores geometry as the dominant term; $T = 0.05$ was rejected because a 0.998 score is effectively a binary accusation. The temperature changes confidence spread, not ranking order. Weights are never tuned to force a scenario's answer — a wrong ranking must be fixed in the geometry or priors.

6.3 Worked example — demo-001

#	Vessel	Type	Overlap	Centre	Course diff.	Match	Score
1	PACIFIC PIONEER	Bulk Carrier	80%	0.9 km	0°	89%	70.6%
2	GOLDEN ENDEAVOUR	Bulk Carrier	24%	4.9 km	1°	58%	20.6%
3	CRIMSON ENDEAVOUR	Chemical Tanker	12%	4.5 km	86°	29%	8.7%

PACIFIC PIONEER ranks first because a discharge simulated from its own track at 2026-01-13 22:00 UTC reproduces 80% of the observed slick, lands 0.9 km from its centre, and is aligned within 0°. The third candidate sailed through the same water but its slick would have been almost perpendicular to the observed one — the orientation term is what separates them.

On the Kutch exercise the top candidate (SAGAR HORIZON) scores 58.3% with a weak 8% overlap — the interface shows that weakness honestly rather than hiding it.

7. Data sources used

Data	Source	Real or simulated
Operational SAR	ISRO EOS-04 L2B ARD, NRSC Bhoonidhi — Gulf of Kutch, 5 Jul 2026 01:17:53 UTC, HH/HV, 18 m	Real (Indian satellite)
Environmental context	MOSDAC Data Download API — INSAT-3DS SST, EOS-06 scatterometer wind, OCM-3 ocean colour (search verified live)	Real catalogue
AIS traffic	NOAA MarineCadastre bulk daily file, Houston Ship Channel (560 vessels)	Real
Training imagery	Deep-SAR SOS (pixel masks) + CSIRO Sentinel-1 binary set (DOI 10.25919/4v55-dn16)	Real, foreign — no labelled Indian set exists
Demo scenes and wind/current	samudra.synth generator with realistic statistics	Simulated
Ground truth for verification	Planted culprit, release time and position, spoofing, loitering, gaps	Simulated by design
Coastline	Natural Earth 1:10 m	Real
Coast Guard stations	Public ICG establishment locations (~1 km)	Real, approximate

The model is trained on open foreign SAR because that is the only labelled oil-spill data that exists; it is run on Indian SAR. The originally planned Zenodo dataset was unavailable due to a documented multi-day Zenodo outage. The brief explicitly permits synthetic AIS; real AIS was used additionally to test against genuine traffic density.

8. Repository and technology stack

Area	Technology
Language / runtime	Python 3.11+ (built on 3.13/3.14)
Data models	Pydantic v2 — every contract in <code>src/samudra/contracts.py</code> , checked against the spec by a test
Geospatial	numpy, pandas, shapely, pyproj (local azimuthal-equidistant projections), rasterio, H3, geopandas
Storage	Parquet + DuckDB (no database server); artifacts on disk
Machine learning	PyTorch / torchvision DeepLabv3+ ResNet-50; weighted cross-entropy + soft Dice loss
API	FastAPI + uvicorn
Frontend	Plain HTML/CSS/JavaScript, Leaflet 1.9.4 (vendored for offline use), no build step
Documents	reportlab (dossier PDF), SHA-256 hash chain
Testing	pytest — text-level UI tests, contract tests, scenario and integrity tests

Key folders

Path	Contents
<code>src/samudra/contracts.py</code>	All data models (Scene, SlickDetection, AisTrack, TrustScore, ReleaseHypothesis, Suspect ...)
<code>src/samudra/synth/</code>	Synthetic scenario generator with ground truth and decoy vessels
<code>src/samudra/ingest/</code>	AIS loader, EOS-04 ingest with self-verification, environment
<code>src/samudra/detection/</code>	Preprocess, segmenter, training, polygonise, fuse
<code>src/samudra/attribution/</code>	<code>drift.py</code> , <code>prune.py</code> , <code>hypothesis.py</code> , <code>rank.py</code> and the CLI entry
<code>src/samudra/trust/</code>	Kinematic, identity, physical and behavioural checks
<code>src/samudra/impact/</code>	Forecast, coastline, land mask
<code>src/samudra/evidence/</code>	PDF dossier and hash chain
<code>src/samudra/dissemination/</code>	Coast Guard alert composition and station list
<code>src/samudra/satellite/</code> , <code>monitor/</code>	Operations console backend: sources, inference, pipeline, polling monitor
<code>src/samudra/api.py</code>	FastAPI server for all dashboards
<code>web/</code>	<code>index.html (/)</code> , <code>dashboard.html (/app)</code> , <code>ops.html (/ops)</code> , <code>investigate/ (/investigate)</code>
<code>config/weights.yaml</code>	All scoring weights, physics constants and thresholds
<code>artifacts/</code>	Pipeline outputs per incident: <code>demo-001</code> , <code>kutch-exercise-001</code> , <code>sat-dropin-...</code>
<code>tests/</code>	pytest suite

Engineering rules the project enforces

- All timestamps UTC; all epoch conversion through `timeutil.epoch_seconds` (pandas 3.0 stores microseconds, so the common `astype(int64)/1e9` idiom is silently 1000× wrong).
- All geometry EPSG:4326 GeoJSON; metric work reprojects locally, never a fixed degrees-per-km constant.
- No pipeline module may read `ground_truth.json` — enforced at runtime and by a source-scanning test.
- No network calls inside `attribution/` or `detection/`; only `ingest` touches the network.
- Every stage is CLI-invokable: `python -m samudra.<module> --incident <id>`.
- Scoring weights are never adjusted to force a correct ranking.

9. The user interfaces

Route	File	Purpose
/	web/index.html	Original single-page investigator view: funnel, conditions, forecast horizons, ranked cards, drift replay, dossier. Unchanged.
/app	web/dashboard.html	Unified timeline from T-16 h hindcast through acquisition to +72 h, animated wind/current from the pipeline's field, Coast Guard alert and dossier.
/ops	web/ops.html	Operations console: Dashboard, Satellite (runs the real model on real SAR and feeds the result into attribution), Spill, Drift, AIS, Capability, Evidence, Help.
/investigate	web/investigate/	New result-first investigator view (section 10).

10. The investigator view redesign (this session)

10.1 Why

The original page was technically complete but hard for a newcomer: the map was dominated by long red trajectory lines, the key result was buried, raw scores such as 0.706 were unexplained, technical terms had no plain-language label, the funnel was not self-explanatory and the legend sat at the bottom of the sidebar. The redesign principle was: **show the result first, then the evidence, then the technical details**.

10.2 How it was done — without breaking anything

- The original `web/index.html` was **copied** to `web/investigate/` and redesigned there; `/`, `/app` and `/ops` are untouched.
- No scoring, drift or forecast is computed in the browser. Every number comes from the API; the only display arithmetic (percentages, `1/area_ratio` as "within N×", forecast-centre displacement between two pipeline polygons) is labelled.
- Three additive API endpoints were added to `api.py`; no existing endpoint changed behaviour:

Endpoint	Why it was needed
GET <code>/investigate</code>	Serves the new page
GET <code>/api/config/scoring</code>	Serves the real weights from <code>config/weights.yaml</code> so the explanation can never drift from the model
GET <code>/api/incident/{id}/replay?mmsi=</code>	Optional mmsi lets Simulate discharge re-run the existing drift model for any candidate, not only #1
GET <code>/api/incident/{id}/scene (+ scene.png)</code>	Serves a stored, georeferenced scene raster and every detected polygon for the Satellite evidence view

10.3 What the new view contains

Area	What it shows
Header	SAMUDRA, 'Simulation mode' badge for synthetic cases, case selector, acquisition time, analysis status, Replay drift, Generate evidence dossier
Case summary	'Oil-like slick detected', location with nearest Coast Guard unit, slick area, estimated age, candidates (3 / 24), conditions, data availability
Pipeline	Detect -> Characterise -> Trace -> Filter -> Attribute -> Forecast, each state derived from real outputs (a scenario-supplied slick shows 'provided', not a tick)
Potential source vessel card	Name, type, MMSI, attribution score as a percentage with an explanation tooltip, margin over the next candidate, key evidence with strength marks, estimated release time and position, View full evidence and Simulate discharge
Map	Orange slick, yellow origin envelope, red selected track (bold only from 2 h before release to acquisition, direction arrows, hourly marks, labelled release point), blue candidates, faint grey filtered traffic, green forecast and cone; on-map legend that doubles as layer toggles; basemap switcher, labels, lat/lon grid, offline coastline
Timeline	Hindcast -> acquisition -> +24/48/72 h; forecasts snap to computed horizons and are never interpolated

Area	What it shows
Discharge simulation	Release point -> particles drifting under the model with the vessel moving -> simulated slick over the observed slick with overlap, centre distance, course alignment and area
Evidence panel	Why the vessel scored X%: each term as a bar with a plain label and its technical value, the scoring model weights, contextual factors as +/-%, and collapsible technical details (formula with real numbers, all hypothesis values, top hypotheses, the model's rationale)
Vessel filtering	Visual funnel with 'Why were they filtered?' listing only the rules <code>prune.py</code> actually applies, plus the filtered vessel list
Compare candidates	Side-by-side table of every candidate with the strongest value per row marked
Satellite evidence	Scene image plus all detections where a georeferenced scene exists; otherwise 'Satellite evidence layer unavailable for this scenario' — no imagery is ever invented
Forecast panel	Predicted area, growth, centre displacement and direction, uncertainty cone, distance to coast, shore contact and ETA
Responsible language	'Potential source vessel', 'attribution score', and the disclaimer that model output does not establish legal responsibility

10.4 Deliberate honesty choices

- 'Data availability: Complete 6/6' is a count of inputs present, not an invented confidence score (no quality metric exists in the pipeline).
- The age cross-check is labelled 'consistent', not 'independent validation', because it depends on an assumed volume.
- The attribution score tooltip explains it is normalised across candidates (sums to 100%) and is not a probability of responsibility.
- '0.9 km' is labelled as the distance between the simulated and observed slick centres, which is what it is.
- Evidence strength bands (Strong ≥ 0.7 , Partial ≥ 0.4) are display bands over the model's own term values, stated as such.
- Wind direction is labelled 'toward', matching the data. (/app, /ops and the alert text say 'from', a 180° labelling error noted for a follow-up fix.)

10.5 Verification performed

Checked in the browser on demo-001 and kutch-exercise-001 at desktop, tablet and phone widths: case switching, forecast horizons, compare dialog (70.6 / 20.6 / 8.7%, 80 / 24 / 12%, 0° / 1° / 86° — matching the data), filter dialog, candidate selection, discharge simulation end to end, drift replay, satellite view (scene available and unavailable), dossier download (HTTP 200, PDF) and no console errors. Two bugs were found and fixed during this work: an animation that could start on a negative first frame, and trust flags (which arrive as objects) rendering as '[object Object]' with a behaviour factor captioned 'No anomalous behaviour flagged' although an AIS gap had raised it. `tests/test_investigate_ui.py` adds 12 tests for the view and its endpoints; the offline suite runs 204 passed, with 7 failures in `test_satellite.py` that need the training archives under `data/raw/oilspill/`, which are not in the repository.

11. Verification and results

11.1 Attribution against planted ground truth

	demo-001	demo-002 (unseen)	houston-001 (real AIS)	kutch-exercise-001
Traffic	Synthetic	Synthetic, 2× density	Real, 560 ships	Real EOS-04 vessel positions
Culprit ranked #1	Yes	Yes	9th of 12 (see below)	Yes
Release time error	0.02 h	0.12 h	0.25 h	—
Release position error	0.98 km	2.34 km	2.53 km	—
Vessels narrowed	24 -> 3	46 -> 3	426 -> 12	164 -> 3
Planted anomalies found	7 / 7	7 / 7	—	—

demo-002 uses a different seed, culprit, twice the traffic and a higher wind band and was never used during development: getting it right is the evidence that the physics generalises rather than constants being fitted. On Houston, no method can name one ship: across all 97 candidates the maximum separation to the nearest moving ship is 72 m, and several tugs run the identical line within the hour, so their simulated slicks are geometrically indistinguishable. The engine still narrows 426 vessels to a shortlist of 12 containing the culprit and pins the release to 15 minutes.

11.2 Components

Component	Result
Drift round-trip	IoU 0.86 against ground truth (gate 0.50), using an independently implemented drift model
CFAR ship detection	13 of 13 planted vessels found at $k = 4.5$; 4 false positives
Trust / behaviour	7 of 7 planted anomalies caught, 0 false positives across 47 ships
Full-scene detection (demo-001)	3 dark features, 3 found; attributed slick IoU 0.841, axis error 0.2°, centroid offset 40 m
Evidence integrity	Altering a file, editing the log or deleting a record are all detected

11.3 Segmentation model (held-out test split, evaluated once)

Metric	Before (weighted CE)	After (CE + Dice)
Oil IoU	0.7191	0.7416
Oil Dice / F1	0.8366	0.8517
Oil precision	0.8279	0.8243
Oil recall	0.8455	0.8809
Mean IoU	0.8042	0.8188
Pixel accuracy	0.9138	0.9199

The training protocol was corrected: the earlier version selected the checkpoint on the test set. Now a deterministic 85/15 train/validation split drives early stopping, checkpoint selection and threshold choice, and the test split is loaded once at the end.

11.4 Detection limit on real EOS-04 HH ocean

Slicks of known area and contrast were planted into real EOS-04 pixels (preserving true speckle) and scored by the real checkpoint, 14 trials per cell:

Area	1 dB	2 dB	3 dB	5 dB	8 dB	12 dB
0.05 km ²	0%	0%	0%	14%	86%	100%
0.10 km ²	0%	0%	21%	93%	100%	100%
0.25 km ²	0%	14%	93%	100%	100%	100%

Area	1 dB	2 dB	3 dB	5 dB	8 dB	12 dB
1.00 km ²	0%	50%	100%	100%	100%	100%

Real oil damps capillary waves by 5-15 dB, so operation sits in the right-hand columns. 0.05 km² (5 hectares) is 200× smaller than the ~10 km² floor a 1-4 km geostationary pixel imposes: small-slick detection is not the weakness — revisit is. False positives over open water: 0.014% of pixels, one region, which was a vessel wake.

12. Honest limitations

- SAR cannot confirm oil; every detection is an oil-like surface anomaly.
- Attribution needs AIS. The real EOS-04 Kutch scene has no AIS on this machine, so it is deliberately not attributed (widening the envelope until some ship falls inside is how innocent vessels get accused).
- The model is trained on VV polarisation and run on EOS-04 HH; the shift is partly quantified by the detection-limit table.
- The baseline has one scene, so anomaly scoring runs in global-fallback mode (needs ≥ 15 per cell).
- prior_offences is always 0 — the repeat-offender database (connector A) is not built.
- The Fay age cross-check is not independent.
- Environment fields in the demo AOIs are synthetic; INSAT-3DS SST and OCM-3 chlorophyll map layers in /app are modelled and badged PROPOSED.
- MOSDAC search is verified live; download and Bhoonidhi retrieval need a registered account and are unverified.
- Alerts are composed, never transmitted. The Coast Guard station list is approximate (~1 km).
- Real-time interception is out of scope: revisit is 5-12 days; the product is forensic attribution and deterrence.

13. How to run it

One-time setup (from the project folder)

```
cd "C:\Users\shari\OneDrive\Desktop\SIH\samudra-sih26143"
python -m venv .venv
.venv\Scripts\python.exe -m pip install -r requirements-demo.txt
.venv\Scripts\python.exe -m pip install -e . --no-deps
```

Start the server

```
.venv\Scripts\python.exe -m samudra.api --port 8000
```

Leave that window open, then open in a browser:

Address	View
http://127.0.0.1:8000/investigate	New investigator view (recommended for demos)
http://127.0.0.1:8000/investigate?case=kutch-exercise-001	Open a specific case
http://127.0.0.1:8000/	Original view
http://127.0.0.1:8000/app	Timeline dashboard
http://127.0.0.1:8000/ops	Operations console

Use 127.0.0.1 rather than localhost (the server is IPv4-only). Stop the server with **Ctrl + C**. If port 8000 is already in use, either close the other server window or use `--port 8001`. Hard-refresh with **Ctrl + F5** if the page looks stale.

Rebuild the pipeline for a case

```
bash scripts/run_demo.sh demo-001 --regen --no-serve
bash scripts/run_demo.sh demo-002 --regen --seed 77341 --vessels 80 --decoys 9 --wind 7.0 9.0
```

Individual stages

```
python -m samudra.synth.generate --scenario demo-001
python -m samudra.attribution --incident demo-001
python -m samudra.trust.score --incident demo-001
python -m samudra.impact.forecast --incident demo-001
python -m samudra.evidence.report --incident demo-001
python -m samudra.evidence.integrity --verify
python -m samudra.dissemination --incident demo-001 --print-message
python -m samudra.satellite --list
```

Tests

```
.venv\Scripts\python.exe -m pytest -q -m "not live"
```

14. Demo script (2-3 minutes)

#	Action	What to say
1	Open /investigate	"An oil-like slick has been detected — here is its size, age and location."
2	Point at the yellow envelope	"We drifted it backwards through wind and current: this is where it could have come from."
3	Point at the funnel	"24 vessels were in the scene; only 3 were inside that envelope at the right time."
4	The source vessel card	"Pacific Pioneer has the highest attribution score, 70.6% — 3.4× the next candidate."
5	Key evidence	"80% of the slick is reproduced, centres 0.9 km apart, aligned with its course."
6	Click Simulate discharge	"We simulate a discharge from its own track and drift it with the same model — it lands on the observed slick."
7	Compare candidates	"The third ship crossed the same water, but its slick would have been 86° off."
8	Click +24 h	"And this is where the slick goes next — 34 km south-east, no shore contact."

#	Action	What to say
9	Satellite evidence	"This is the scene and every dark feature the detector found."
10	Generate evidence dossier	"A hash-chained PDF for the Coast Guard. It supports a decision; it does not establish legal responsibility."

15. Glossary

Term	Meaning
SAR	Synthetic-aperture radar — sees surface roughness day and night, through cloud
EOS-04	ISRO's C-band SAR satellite (RISAT-1A), the sensor the brief names
AIS	Automatic Identification System — unencrypted VHF self-reports of ship position, speed and identity
MMSI	Maritime Mobile Service Identity — a vessel's AIS number
Hindcast	Running the drift model backwards in time
Origin envelope	The space-time region the oil could have come from
IoU	Intersection over union — overlap of two shapes (1 = identical)
Posterior / attribution score	Relative ranking score across candidates after priors, summing to 100%
Prior	A contextual multiplier (vessel type, AIS trust, behaviour, gaps)
CFAR	Constant false-alarm rate — radar ship-detection method
NOS-DCP	National Oil Spill Disaster Contingency Plan — Tier I local, II regional, III national
MARPOL Annex I	International regulations for the prevention of pollution by oil
Hash chain	Each audit entry includes the hash of the previous one, so any alteration is detectable

16. What remains to be built

Item	Value
Repeat-offender database (connector A)	Closes the feedback loop so the system compounds across incidents
SAR-to-AIS matching (vessels/match.py)	Enables DARK and PHANTOM classification — the strongest anti-spoofing signal
15+ scenes per AOI	Activates the per-cell baseline memory
Credentialed MOSDAC / Bhoonidhi retrieval	Automatic ingest of real environmental fields and SAR
Optical branch (Sentinel-2/3 equivalent)	Tighter time brackets from multi-sensor passes
Wind label fix in /app, /ops and alert text	Correct 'from' to 'toward'

SAMUDRA's claim is specific and verifiable: the attribution engine works, and its answers can be checked against planted ground truth. Everything real and everything simulated is labelled as such.